Quantum Mechanics II — Exercise Sheet 14

The Coulomb Hamiltonian and stability

Variant: Questions only

Exercise 1 (A Gaussian trial energy). Let $H = -\hbar^2\Delta/(2m) - C/r$ in three dimensions, where $C > 0$. For $b > 0$ take the normalized Gaussian $\psi_b(x) = (\pi b^2)^{-3/4} e^{-r^2/(2b^2)}$. (i) Compute $\langle H\rangle_b$ and minimize it over b. (ii) Compare the minimum with the exact lower bound $-\mathrm{Ry} = -mC^2/(2\hbar^2)$.

Exercise 2 (Reduced mass and isotope shifts). A nucleus of mass $M > 0$ and a particle of mass $m_e > 0$ interact through $-C/r$ in their relative coordinate. Take the separation of centre-of-mass motion as given, with reduced mass $m_{\mathrm{red}} = m_e M/(m_e + M)$. Find the ratios of the Bohr length and Rydberg energy to their values at an infinitely heavy nucleus. Expand each ratio through first order in m_e/M.

Exercise 3 (A virial check on an eigenstate). A normalized bound Coulomb eigenstate has energy $E < 0$ and satisfies the lecture's virial identity. (i) Find its kinetic and potential means and $\langle r^{-1}\rangle$ in terms of E and C. (ii) Then use Cauchy–Schwarz to bound $\langle r\rangle$ whenever that mean is finite. Do not assume the hydrogen level formula from Lecture 15.

Exercise 4 (Scaling the domain and the energy). For $a > 0$ let $(D_a f)(x) = a^{3/2} f(ax)$ and $H_C = -\hbar^2\Delta/(2m) - C/r$ on the Coulomb domain $D(H_0)$. (i) Show that D_a is unitary and preserves $D(H_0)$. (ii) Compute $D_a^{-1} H_C D_a$ and express it using $H_{C/a}$. Deduce how the bottom energy scales when C is multiplied by a.